

Key success factors

1. Successful penetration into Asia-Pacific's emerging markets
2. Expansion to global markets and creation of strong brand recognition through licensing
3. Ability to innovate around-the-box and engage consumers through customer feedback and crossover toys

offers a plethora of potential for companies to pursue. But the interests of children, who are the primary consumers for toys, are changing faster than ever. Toy manufacturers have to create innovative toys to capture their interests.

Toys these days are popular not only among kids, but adults too have entered this field through the medium of sports and games. Many sports and games are played by adults at national and international levels representing their respective countries. Parents too prefer to play with their children in order to interact with the latter and thus engage in board games and other toys with them. The metros and category A towns account for most branded purchases and those that sell at higher price points. Largely, the rest of the market is highly price sensitive and items above ₹ 200 fall in the planned purchase category and not impulse buying. In category C and D towns, unbranded and lower-priced toys are sold at average price points of below ₹ 100. However, the scene in India is changing very fast and there is enough scope for more number of players in the field.

Raw materials

Some common examples of electronic toys are remote control cars, model trains and racing car sets, educational toys such as toy computers, radio-controlled aeroplanes and drones, LED toys and so on. There are differences in the mechanical assembly of these types of toys, but the basic electronics principle is the same. These types of toys have four main units.

Transmitter. The transmitter sends radio waves to the receiver.

Receiver. An antenna and circuit board inside the toy receives signals from the transmitter and activates motors inside the toy as commanded by the transmitter.

TABLE I FIXED CAPITAL			
LAND AND BUILDING			
Description	Details		
Built-up area	200 sqm		
Office/stores	40 sqm		
Factory	160 sqm		
Rent (₹ per month)	10,000		
PLANT, MACHINERY AND EQUIPMENT			
Description	Indian/Imported	Quantity	Cost (₹)
4½-digit digital multimeter	Indian	2	22,000
Temperature-controlled soldering unit	Indian	4	20,000
LCR meter	Indian	2	20,000
Drilling machine	Indian	1	6000
Analogue multimeter	Indian	2	2000
Toolkit	Indian	4	20,000
Electronic screw driver and screw feeder	Indian	5	30,000
Combined soldering de-soldering station	Indian	2	35,000
High-speed mini drill set	Indian	3	30,000
60MHz digital storage oscilloscope	Indian	1	60,000
Personal computer with UPS and printer	Indian	2	80,000
Electrification charges @ 10% of machinery and equipment			32,500
Office furniture, working tables and equipment			50,000
Tool, dies and equipment			50,000
Pre-operative expenses			50,000
Total Fixed Capital			507,500

Motor(s). The transmitter sends a control signal to the receiver using radio waves, which then drives a motor, causing a specific action to occur. The motor(s) in a toy car may cause the wheels to turn, steer the vehicle, operate propellers and the like.

Power source. This is typically a rechargeable battery pack, but sometimes it is just regular battery cells.

The manufacturing process involves assembly of electronic circuits, electromechanical hardware parts, mechanical assembly and other sub-assembly parts as per the design. Subsequently, the electronics assembly—ICs, transistor, diodes, resistors, capacitors, coils and electromagnetic relays—are assembled on PCBs as per requirement. The assembled PCBs are tested for the desired performance. The electronics assembly along with the electromechanical assembly, hardware such as connectors/switches, mechanical assembly and LEDs are assembled and housed in a fibre/plastic toy car case.

Waste management

Plastic forms a significant part of the toy industry. Even electronic toys use plastic in their manufacturing process, either in the console or for packaging and product itself, and most are not recyclable. The government places utmost importance on curbing environmental pollution, thus making it necessary for producers of electronic toys to use environment-friendly techniques of production and adopt pollution-control measures by process modification and technology substitution. To curb emission of harmful pollutants into the atmosphere, the following steps can be taken.

In the electronics industry, fumes and gases are released during soldering, which are harmful to people as well as the environment. Alternate eco-friendly technologies may be used to phase out existing manufacturing technologies. Numerous new fluxes have been developed containing two to ten per cent solids as opposed to